

Tinghui Li

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EDUCATION

University of Sydney Sydney, Australia
Doctor of Philosophy 10/2023 – 03/2027

- **Supervisors:** Dr. Zhanna Sarsenbayeva, A/Prof. Anusha Withana, Prof. Eduardo Velloso
- **Research Area:** Human-Computer Interaction, Extended Reality, Accessibility
- **Project Topic:** Understanding and Modelling the Effects of Situational-Induced Impairments and Disabilities in Extended Reality
- **Scholarships:** Tuition Fee Scholarship, Stipend Scholarship

University of Sydney Sydney, Australia
Bachelor of Advanced Studies (Honours) 09/2022 – 09/2023

- **Major:** Computer Science (First Class Honours)
- **Supervisor:** A/Prof. Anusha Withana
- **Project Topic:** Physics Inspired Time-Agnostic Graph Neural Network for Finger Motion Prediction

University of Sydney Sydney, Australia
Bachelor of Science 02/2020 – 09/2022

- **Majors:** Data Science, Biochemistry and Molecular Biology
- **Courses:** Human-in-the-Loop Data Analytics (91/100); Pervasive Computing (91/100); Introduction to Programming (89/100); Visual Analytics (88/100); Usability Engineering (82/100).

EXPERIENCE

School of Information Technology, University of Sydney Sydney, Australia
Casual Academic – Computer Science 06/2023 – Present

- INFO3315 Human-Computer Interaction Teaching Assistant and Tutor
- COMP4427/5427 Usability Engineering Teaching Assistant and Tutor
- COMP5703/DATA5703/CSEC5703 Group-Based Capstone Project Tutor

Bosch (China) Investment Co., Ltd. Shanghai, China
Data Scientist Intern 11/2021 – 01/2022

- Responsible for daily data transmission, data cleaning and data proofreading;
- Analyzed the abnormal vehicles data, including finding the abnormal cells, plotting current and voltage graphs, and finding the difference between normal and abnormal vehicles;
- Participated in the production and improvement of the PowerBI dashboard.

University College London London, England
Exchange Summer School 07/2021 – 08/2021

- **Courses:** Health Data Science and Data Analytics in Healthcare.

Shanghai Jiao Tong University Shanghai, China
Exchange Summer School 12/2020 – 01/2021

- **Courses:** Algorithms and Analysis (89/100); Statistics of Stochastic Processes (89/100).

Faculty of Science, University of Sydney Sydney, Australia
Peer Mentor 12/2020 – 06/2021

- Provided practical information about the course, faculty/school and university life;
- Directed new students to relevant support services and resources;
- Shared my knowledge and experiences as a student.

PUBLICATIONS

- [CHI '25] **Tinghui Li**, Eduardo Velloso, Anusha Withana, & Zhanna Sarsenbayeva. Estimating the Effects of Encumbrance and Walking on Mixed Reality Interaction. CHI Conference on Human Factors in Computing Systems.
- [Under Review] **Tinghui Li**, Eduardo Velloso, Anusha Withana, & Zhanna Sarsenbayeva. WICE: A Model to Quantify Arm Fatigue when Carrying Objects across Different Arm Locations.
- [Under Review] **Tinghui Li**, Niels van Berkel, Brandon Victor Syiem, Anusha Withana, & Zhanna Sarsenbayeva. Trends and Directions of Extended Reality Research in HCI: A Co-Word Analysis.
- [Under Review] **Tinghui Li**, Pamuditha Somarathne, Zhanna Sarsenbayeva, & Anusha Withana. Never-Lagging: Enhancing Virtual Reality Finger Tracking with a Physics-Inspired Time-Agnostic Graph Neural Network.
- [Under Review] Yingqi Wang, **Tinghui Li**, & Zhanna Sarsenbayeva. Exploring Virtual Reality Interaction in Confined Spaces: A Third-Person Perspective Solution.
- [Under Review] Jens Emil Sloth Grønbaek, Adélaïde Genay, Andrew Irlitti, Brandon Victor Syiem, Jarrod Knibbe, Emily Wong, **Tinghui Li**, Zhongyi Bai, & Eduardo Velloso. Distributed Mixed Reality Collaboration: From Vision to Mission.
- [Ongoing] **Tinghui Li**, Danyang Peng, & Zhanna Sarsenbayeva. Estimating the Effects of Noise on Mixed Reality Interaction.
- [Ongoing] **Tinghui Li**, Vittoria Salim, & Zhanna Sarsenbayeva. Optimizing VR Safety in Confined Spaces with AI Driven Small Objects Boundary System.

AWARDS

- 2025 University of Sydney Postgraduate Research Support Scheme Funding (A\$2,500)
- 2023 – 2027 University of Sydney Faculty of Engineering Research Stipend Scholarship (A\$40,109 p.a.)
- 2023 – 2027 University of Sydney International Tuition Fee Scholarship
- 2023 University of Sydney Engineering Vacation Research Internship Program Scholarship (A\$3,000)
- 2023 University of Sydney Undergraduate High Honour Roll
- 2022 University of Sydney Undergraduate Coursework and Top 3rd Year High Honour Roll
- 2021 University of Sydney Global Citizenship Award

SUPERVISION

- **Ruilin Liu**: 2025 Bachelor of Advanced Studies (Honours), USYD - Co-supervisor
- **Vittoria Salim**: 2024 Bachelor of Advanced Computing (First-Class Honours), USYD - Co-supervisor
- **Yingqi Wang**: 2024 Bachelor of Advanced Computing (First-Class Honours), USYD - Co-supervisor
- **Jennifer Lai**: 2023 Summer Vacation Research Intern, USYD - Co-supervisor

ACTIVITIES

- [Reviewer] The ACM CHI conference on Human Factors in Computing Systems (CHI '25)
- [Reviewer] 36th Australian Conference on Human-Computer Interaction (OzCHI '24)
- [Reviewer] The Augmented Humans (AHs) International Conference (AHs '24)
- [Student Volunteer] 22nd IEEE International Symposium on Mixed and Augmented Reality (ISMAR '23)
- [Student Volunteer] 48th International Conference on Very Large Databases (VLDB '22)

SKILLS

Technical Skills

- Proficient in Python, R, C#, SQL, and LaTeX.
- Proficient in using Unity, Oculus, Figma, Tableau, and PowerBI.
- Know about PyMOL, PeptideMass, BLAST, ProtParam, Uniprot.
- Know about Blender, C/C++, and MATLAB.

Languages

- Mandarin (Native Speaker)
- English (Proficient, IELTS 6.5, Writing 7.0)